

The Generalized Regression Model and Heteroskedasticity

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Intro

We now extend the multiple regression model to incorporate disturbances that violate assumption A.4.

But first things first: is the correct name *heteroskedasticity* or *heteroscedasticity*?

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MISCELLANEA

ON HETEROS*EDASTICITY

By J. HUSTON McCULLOCH¹

THE MOST PRESSING ISSUE in econometric orthography today is whether heteros*edasticity should be spelled with a k or with a c. Heteroskedasticity is used in their texts by Dhrymes (1970), Goldberger (1964), Intriligator (1978), Kmenta (1971) and Valavanis (1959), while heteroscedasticity is preferred by Champenowse (1969), Chow (1983), Goldfeld and Quandt (1977), Johnston (1963), Maddala (1979), Malinvaud (1970), and Theil (1971).²

Our word is a $\alpha\alpha$ -adern coinage, derived from the two Greek roots hetero-($\epsilon\tau\epsilon\rho\alpha$), meaning "other" or "different," and skedannumi ($\sigma\kappa\epsilon\delta\alpha\iota\nu\mu\iota$), meaning "to scatter." The letter in question is therefore the transliteration of Greek kappa (κ).

In scientific words which scholars have lifted directly from Greek into English, the letter kappa is always transliterated as k. Examples are skeptic ($\sigma\kappa\epsilon\pi\tau\iota\kappa\acute{o}\varsigma$) and skeleton ($\sigma\kappa\epsilon\lambda\epsilon\tau\epsilon\rho\acute{o}\varsigma$).

Greek kappa does sometimes make its way into English as c, but only in common words which entered English through French and old scientific words that entered through Latin. Examples are sceptre ($\sigma\kappa\epsilon\pi\tau\rho\acute{\nu}\varsigma$), scene ($\sigma\kappa\eta\eta\acute{\nu}\eta$) and cyclic ($\kappa\iota\kappa\lambda\iota\kappa\acute{o}\varsigma$). Kappa becomes c in French or Latin, simply because k is not used in these languages except to spell foreign proper names. When such a c is followed by e, i, or y, however, it is always sibilant. The only way a kappa taken into French can retain its "k" sound before one of these vowels is in the rare event that it becomes "qu" (as in *squelette*).

In English as in French and Latin, c before e is always soft.³ Examples include ceiling, celerey, ceremony, cease, cedar, celestial, celibacy, cell, cement, cent, center, necessary, scent, etc., any of which would sound ridiculous with a hard c.

If heteros*edasticity were spelled with a c, it would thus have had to have entered the English language either in 1066 with the Norman invaders or else in the middle ages from Latin, neither of which was the case. Furthermore, it would have to be pronounced "heteroscedasticity," which it is not.

Heteroskedasticity is therefore the proper English spelling.⁴

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¹ The author is indebted to Evangelos Falaris and Jerry Thursby for invaluable technical assistance. Any errors or omissions remain the responsibility of the author.

² The earliest use of either I could find was in a 1923 statistics text by Truman L. Kelley. Kelley (as well as Kendall and Stuart, 1958) uses "homoscedastic" and would presumably prefer "heteroscedastic." (Tests cited are not listed, at editor's request.)

³ An alleged exception to this rule is Celtic, which is sometimes mispronounced "keltic" in imitation of German *Keltisch*. This word derives from Greek $\kappa\epsilon\lambda\tau\iota\kappa\acute{o}\varsigma$. Since the English spelling is ordinarily Celtic, not Keltic, it has entered through French (where *celtique* is pronounced with soft c) and not directly from Greek or even indirectly through German as occasionally happens with Greek words. Boston basketball fans therefore pronounce this word correctly, while afflicted academics are in error, unless they actually spell it with a k.

⁴ *Hétéroskedasticité* is correct in French (and must be pronounced with a soft c), but has nothing to do with English, which has a k to transliterate kappa in scientific terms derived from Greek. *Cinéma* (short for *cinématographe*, from $\kappa\iota\gamma\epsilon\mu\alpha$ + $\gamma\omega\phi\acute{o}\varsigma$) was already a household word in French before it entered English and therefore retains its (soft) c.

Introduction

The **Generalized Linear Regression Model** is

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon$$

$$E[\varepsilon|\mathbf{X}] = \mathbf{0}$$

$$E[\varepsilon\varepsilon'|\mathbf{X}] = \sigma^2\mathbf{\Omega} = \mathbf{\Sigma}$$

where $\mathbf{\Omega}$ is a positive definite matrix.

Notice that $\sigma^2\mathbf{I}$ is a special case.

Introduction

The two leading cases are **heteroskedasticity** and **autocorrelation**.

Heteroskedasticity in cross-sectional data appear in data where the scale of the dependent variable and the explanatory power of the model tend to vary across observations.

- For example, we observe greater variation in expenditure on certain commodity groups among high-income family than in low-income ones.

Let's focus on this issue now, still assuming that disturbances are uncorrelated across observations.

Introduction

In the current context, $\sigma^2\mathbf{\Omega}$ would be

$$\sigma^2\mathbf{\Omega} = \sigma^2 \begin{bmatrix} \omega_1 & 0 & \cdots & 0 \\ 0 & \omega_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \omega_n \end{bmatrix} = \begin{bmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_n^2 \end{bmatrix}$$

Robust Least Squares Estimation and Inference

The generalized model drops assumption A.4., allowing disturbances to being heteroskedastic or autocorrelated (or both).

The Least Squares estimator is said to be “robust” if it is insensitive to departures from the base assumptions of the model.

We can distinguish between

1. **Robust estimator:** A consistent estimator that remain consistent in spite of violations of assumptions used to motivate it.
2. **Robust inference:** Inference procedures are robust when they are based on estimators of asymptotic variances that are appropriate even when the assumptions are violated.

A Heteroskedasticity-Robust Covariance Matrix for Least Squares

For the most general case, suppose $Var[\varepsilon_i|\mathbf{x}_i] = \sigma_i^2$

We can write the LS estimator as

$$\mathbf{b} = \beta + (\mathbf{X}'\mathbf{X})^{-1} \sum_{i=1}^n \mathbf{x}_i \varepsilon_i$$

Then,

$$Var[\mathbf{b}|\mathbf{X}] = \frac{1}{n} \left(\frac{\mathbf{X}'\mathbf{X}}{n} \right)^{-1} \left(\frac{\mathbf{X}'(\sigma^2\mathbf{\Omega})\mathbf{X}}{n} \right) \left(\frac{\mathbf{X}'\mathbf{X}}{n} \right)^{-1}$$

The asymptotic variance will be

$$Asy.Var[\mathbf{b}] = \frac{1}{n} \mathbf{Q}^{-1} \left[\text{plim} \frac{1}{n} \sum_{i=1}^n \sigma_i^2 \mathbf{x}_i \mathbf{x}_i' \right] \mathbf{Q}^{-1} = \frac{1}{n} \mathbf{Q}^{-1} \mathbf{Q}^* \mathbf{Q}^{-1}$$

A Heteroskedasticity-Robust Covariance Matrix for Least Squares

Two points to consider:

1. Is $s^2(\mathbf{X}'\mathbf{X})^{-1}$ likely to be a valid estimator of $Asy.Var[\mathbf{b}]$?
2. If not, is there a strategy available that is “robust” to unspecified heteroskedasticity?

Based on the expression for $Var[\mathbf{b}|\mathbf{X}]$, we see that $s^2(\mathbf{X}'\mathbf{X})^{-1}$ would not be the appropriate estimator for the asymptotic covariance matrix for the LS estimator.

The answer to the second question is yes. What we need is a feasible estimator of $\mathbf{Q}^*$.

White's (1980) Heteroskedasticity-Robust Estimator

White's (1980) Heteroskedasticity-Robust Estimator of $\mathbf{Q}^*$ is

$$\mathbf{W}_{het} = \frac{1}{n} \sum_{i=1}^n e_i^2 \mathbf{x}_i \mathbf{x}_i',$$

where e_i is the LS residual, $y_i - \mathbf{x}_i' \mathbf{b}$.

Thus, we have

$$Est.Asy.Var[\mathbf{b}] = n(\mathbf{X}'\mathbf{X})^{-1} \mathbf{W}_{het} (\mathbf{X}'\mathbf{X})^{-1}$$

The implication is that we can discard the homoskedasticity assumption in A.4. and recover appropriate standard errors.

Robustness to Clustering

Settings in which the sample data consists of groups of related observations are increasingly common:

1. firms grouped by industries
2. students in schools
3. homes in neighborhoods

Suppose that the sample consists of C groups of observations, labeled $c = 1, \dots, C$. There are N_c observations in group (or cluster) c , with $c \geq 1$.

- Thus, $\sum_c n$.

Robustness to Clustering

The regression model is

$$y_{i,c} = \mathbf{x}'_{i,c}\beta + \varepsilon_{i,c}$$

The LS estimator is

$$\mathbf{b} = \beta + \left(\sum_{c=1}^C \mathbf{X}'_c \mathbf{X}_c \right)^{-1} \left[\sum_{c=1}^C \left(\sum_{i=1}^N \mathbf{x}_{ii,c} \varepsilon_{i,c} \right) \right] = \beta + (\mathbf{X}'\mathbf{X})^{-1} \left[\sum_{c=1}^C \mathbf{X}'_c \varepsilon_c \right]$$

where $\mathbf{X}_c$ is the $N_c \times K$ matrix of exogenous variables for cluster c and ε_c is the vector of N_c disturbances for the group.

Here we assume that observations within a cluster are correlated but the clusters are independent.

Robustness to Clustering

The variance matrix of the OLS estimator which is robust to clustering is

$$Var[\mathbf{b}|\mathbf{X}] = (\mathbf{X}'\mathbf{X})^{-1} \left[\sum_{c=1}^C X_c \boldsymbol{\Omega}_c X_c' \right] (\mathbf{X}'\mathbf{X})^{-1}$$

The construction is essentially the same as in the White estimator, though the matrix $\boldsymbol{\Omega}_c$ is the matrix of variances and covariances for the full vector ε_c .

- It would be identical to the White estimator if each cluster contained one observation.

The asymptotic covariance matrix is

$$Asy.Var[\mathbf{b}] = \frac{1}{n} \mathbf{Q}^{-1} \left[\text{plim} \sum_{c=1}^C X_c \boldsymbol{\Omega}_c X_c' \right] \mathbf{Q}^{-1}$$

The practical issue is that the asymptotic result holds only as the number of clusters C increases. In practice, C is rather small.

Robustness to Clustering

Like before, we need a feasible estimator for the term in brackets.

We can use

$$\mathbf{W}_{\text{cluster}} = \frac{1}{C} \sum_{c=1}^C (\mathbf{X}'_c \mathbf{e}_c) (\mathbf{X}'_c \mathbf{e}_c)' = \frac{1}{C} \sum_{c=1}^C \left(\sum_{i=1}^{N_c} \mathbf{x}_{ic} \mathbf{e}_{ic} \right) \left(\sum_{i=1}^{N_c} \mathbf{x}_{ic} \mathbf{e}_{ic} \right)'$$

Then,

$$Est.Var.Asy[\mathbf{b}] = \mathbf{C}(\mathbf{X}'\mathbf{X})^{-1} \mathbf{W}_{\text{cluster}} (\mathbf{X}'\mathbf{X})^{-1}$$

Bootstrapped Standard Errors with Clustered Data

The preceding discussion assumes that the sample consists of a reasonably large number of clusters, drawn randomly from a very large population of clusters.

The method used to estimate the covariance matrix involves using the data in $\mathbf{X}$ and the LS residuals.

Bootstrapping is another method that is likely to be effective under those assumed sampling conditions.

Bootstrapped Standard Errors with Clustered Data

The basic steps in the methodology are:

1. For R repetitions, draw a random sample of N_c observations from the full sample of N_c observations with **replacement**. Estimate the parameters of the regression model with each of the R constructed samples.
2. The estimator of the asymptotic covariance matrix is the sample variance of the R sets of estimated coefficients.

Robust Covariance Matrix Estimation

To sum up, covariance-robust estimators allow us to still make appropriate inferences based on the LS estimators without actually specifying the type of heteroskedasticity involved.

Note: The asymptotic properties of the estimator are unambiguous, but the results in small samples can be problematic:

- squared residuals tend to underestimate the squares of the true disturbances (we need to make adjustments in such cases).

Properties of the Least Squares

We now consider the generalized model outlined in the introduction:

$$\begin{aligned}\mathbf{y} &= \mathbf{X}\beta + \varepsilon \\ E[\varepsilon|\mathbf{X}] &= \mathbf{0} \\ E[\varepsilon\varepsilon'|\mathbf{X}] &= \sigma^2\mathbf{\Omega} = \mathbf{\Sigma}\end{aligned}$$

where $\mathbf{\Omega}$ is a positive definite matrix.

Here we aim to analyze which of the properties of the classical model still hold.

Finite sample Properties of the Least Squares

Under the generalized model, we have

$$E[\mathbf{b}|\mathbf{X}] = \beta \Rightarrow E[\mathbf{b}] = \beta$$

and

$$Var[\mathbf{b}|\mathbf{X}] = \frac{1}{n} \left(\frac{\mathbf{X}'\mathbf{X}}{n} \right)^{-1} \left(\frac{\mathbf{X}'(\sigma^2\mathbf{\Omega})\mathbf{X}}{n} \right) \left(\frac{\mathbf{X}'\mathbf{X}}{n} \right)^{-1}$$

Because the variance of the LS estimator is not $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$, statistical inference based on $s^2(\mathbf{X}'\mathbf{X})^{-1}$ may be misleading.

Finite sample Properties of the Least Squares

THEOREM 9.1 Finite-Sample Properties of $\mathbf{b}$ in the Generalized Regression Model

If the regressors and disturbances are uncorrelated, then the least squares estimator is unbiased in the generalized regression model. With nonstochastic regressors, or conditional on $\mathbf{X}$, the sampling variance of the least squares estimator is given by (9-8). If the regressors are stochastic, then the unconditional variance is $E_{\mathbf{X}} [\text{Var}[\mathbf{b}|\mathbf{X}]]$. From (9-6), $\mathbf{b}$ is a linear function of $\boldsymbol{\varepsilon}$. Therefore, if $\boldsymbol{\varepsilon}$ is normally distributed, then $\mathbf{b}|\mathbf{X} \sim N[\boldsymbol{\beta}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\boldsymbol{\Omega}\mathbf{X})(\mathbf{X}'\mathbf{X})^{-1}]$.

Asymptotic Properties of the Least Squares

In the heteroskedastic case, if the variances of ε_i are finite and are not dominated by a single term, so that the conditions of the Lindeberg-Feller CLT applies, then the LS estimator is asymptotically normally distributed with covariance matrix

$$Asy.Var[\mathbf{b}] = \frac{1}{n} \mathbf{Q}^{-1} \left[\text{plim} \frac{1}{n} \sum_{i=1}^n \sigma_i^2 \mathbf{x}_i \mathbf{x}_i' \right] \mathbf{Q}^{-1}$$

For the most general case, asymptotic normality is much more difficult to establish because we might not necessarily have sums of independent or even uncorrelated random variables.

Nonetheless, Amemiya (1985) and Anderson (1971) have established the asymptotic normality of $\mathbf{b}$ in a model of autocorrelated disturbances general enough to include most of the settings we are likely to meet in practice.

Asymptotic Properties of the Least Squares

THEOREM 9.2 Asymptotic Properties of $\mathbf{b}$ in the Generalized Regression Model

If $\mathbf{Q} = \text{plim}(\mathbf{X}'\mathbf{X}/n)$ and $\text{plim}(\mathbf{X}'\mathbf{\Omega}\mathbf{X}/n)$ are both finite positive definite matrices, then $\mathbf{b}$ is consistent for $\boldsymbol{\beta}$. Under the assumed conditions, $\text{plim } \mathbf{b} = \boldsymbol{\beta}$. If the regressors are sufficiently well behaved and the off-diagonal terms in $\mathbf{\Omega}$ diminish sufficiently rapidly, then the least squares estimator is asymptotically normally distributed with mean $\boldsymbol{\beta}$ and asymptotic covariance matrix given in (9-12).

Heteroskedasticity and $Var[\mathbf{b}|\mathbf{X}]$

We have that, for well-behaved regressors,

$$Asy.Var[\mathbf{b}|\mathbf{X}] = \frac{\sigma^2}{n} \mathbf{Q}^{-1} \left(\text{plim} \sum_{i=1}^n \omega_i \mathbf{x}_i \mathbf{x}_i' \right) \mathbf{Q}^{-1}$$

In general, we have

$$\mathbf{b} \sim_a N\left[\beta, \frac{\sigma^2}{n} \mathbf{Q}^{-1} \mathbf{Q}^* \mathbf{Q}^{-1}\right], \text{ with } \mathbf{Q}^* = \text{plim} \sum_{i=1}^n \omega_i \mathbf{x}_i \mathbf{x}_i'$$

The conventionally estimated covariance matrix for the LS estimator, however, is $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$, which is inappropriate.

The appropriate matrix is $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{\Omega}\mathbf{X})(\mathbf{X}'\mathbf{X})^{-1}$

Heteroskedasticity and $Var[\mathbf{b}|\mathbf{X}]$

We now consider how erroneous the conventional estimator is likely to be. The difference is

$$Est.Var[\mathbf{b}|\mathbf{X}] - Var[\mathbf{b}|\mathbf{X}] = s^2(\mathbf{X}'\mathbf{X})^{-1} - \sigma^2(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{\Omega}\mathbf{X})(\mathbf{X}'\mathbf{X})^{-1}$$

In large samples, where $s^2 \approx \sigma^2$, the difference is approximately equal to

$$\mathbf{D} = \frac{\sigma^2}{n} \left(\frac{\mathbf{X}'\mathbf{X}}{n} \right)^{-1} \left[\frac{\mathbf{X}'\mathbf{X}}{n} - \frac{\mathbf{X}'\mathbf{\Omega}\mathbf{X}}{n} \right] \left(\frac{\mathbf{X}'\mathbf{X}}{n} \right)^{-1}$$

Heteroskedasticity and $Var[\mathbf{b}|\mathbf{X}]$

The difference between the two matrices hinges on the bracketed matrix,

$$\Delta = \sum_{i=1}^n (1/n) \mathbf{x}_i \mathbf{x}_i' - \sum_{i=1}^n (\omega_i/n) \mathbf{x}_i \mathbf{x}_i' = \frac{1}{n} \sum_{i=1}^n (1 - \omega_i) \mathbf{x}_i \mathbf{x}_i'$$

These are two weighted averages of the matrices $\mathbf{x}_i \mathbf{x}_i'$ using weights 1 for the first term and ω_i for the second.

Efficient Estimation by Generalized Least Squares (GLS)

Efficient estimation of β in the generalized regression model requires knowledge of Ω .

Let's consider the case in which Ω is a known, symmetric, positive definite matrix.

Because Ω is a positive definite symmetric matrix, it can be factored into

$$\Omega = \mathbf{C}\Lambda\mathbf{C}',$$

where the columns of $\mathbf{C}$ are the characteristic vectors of Ω and the characteristic roots of Ω are arrayed in the diagonal matrix Λ .

Generalized Least Squares (GLS)

Define $\mathbf{\Lambda}^{1/2}$ as the diagonal matrix with the i th diagonal element $\sqrt{\lambda_i}$ and let $\mathbf{T} = \mathbf{C}\mathbf{\Lambda}^{1/2}$. Then,

$$\mathbf{T}\mathbf{T}' = \mathbf{\Omega}$$

Also let $\mathbf{P}' = \mathbf{C}\mathbf{\Lambda}^{-1/2}$, so $\mathbf{\Omega}^{-1} = \mathbf{P}'\mathbf{P}$

If we pre-multiply $\mathbf{y} = \mathbf{X}\beta + \varepsilon$ by $\mathbf{P}$, we obtain

$$\mathbf{P}\mathbf{y} = \mathbf{P}\mathbf{X}\beta + \mathbf{P}\varepsilon$$

$$\mathbf{y}_* = \mathbf{X}_*\beta + \varepsilon_*$$

Generalized Least Squares (GLS)

The conditional variance of ε_* is

$$E[\varepsilon_* \varepsilon_*' | \mathbf{X}] = E[\mathbf{P} \varepsilon \varepsilon' \mathbf{P}' | \mathbf{P} \mathbf{X}] = \mathbf{P} \sigma^2 \boldsymbol{\Omega} \mathbf{P}' = \sigma^2 \mathbf{I}$$

so the classical regression model applies to the transformed model.

Since it is assumed that $\boldsymbol{\Omega}$ is known, we can define

$$\begin{aligned}\hat{\beta} &= (\mathbf{X}_*'\mathbf{X}_*)^{-1}\mathbf{X}_*'\mathbf{y}_* \\ &= (\mathbf{X}'\mathbf{P}'\mathbf{P}\mathbf{X})^{-1}\mathbf{X}'\mathbf{P}'\mathbf{P}\mathbf{y} \\ &= (\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{y}\end{aligned}$$

Generalized Least Squares (GLS)

We have that $\hat{\beta}$ is the efficient estimator of β and it is called the Generalized Least Squares (GLS)

This is in contrast to the OLS estimator, which uses $\mathbf{I}$ as the weighting matrix instead of $\mathbf{\Omega}^{-1}$.

What can we say about the properties of the GLS estimator compared to the OLS?

Properties of the Generalized Least Squares

THEOREM 9.4 Properties of the Generalized Least Squares Estimator

If $E[\boldsymbol{\varepsilon}_* | \mathbf{X}_*] = \mathbf{0}$, then

$$E[\hat{\boldsymbol{\beta}} | \mathbf{X}_*] = E[(\mathbf{X}_*'\mathbf{X}_*)^{-1}\mathbf{X}_*'\mathbf{y}_* | \mathbf{X}_*] = \boldsymbol{\beta} + E[(\mathbf{X}_*'\mathbf{X}_*)^{-1}\mathbf{X}_*'\boldsymbol{\varepsilon}_* | \mathbf{X}_*] = \boldsymbol{\beta}.$$

The GLS estimator $\hat{\boldsymbol{\beta}}$ is unbiased. This result is equivalent to $E[\mathbf{P}\boldsymbol{\varepsilon} | \mathbf{P}\mathbf{X}] = \mathbf{0}$, but because $\mathbf{P}$ is a matrix of known constants, we return to the familiar requirement $E[\boldsymbol{\varepsilon} | \mathbf{X}] = \mathbf{0}$. The requirement that the regressors and disturbances be uncorrelated is unchanged.

The GLS estimator is consistent if $\text{plim}(1/n)\mathbf{X}_*'\mathbf{X}_* = \mathbf{Q}_*$, where $\mathbf{Q}_*$ is a finite positive definite matrix. Making the substitution, we see that this implies

$$\text{plim}[(1/n)\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{X}]^{-1} = \mathbf{Q}_*^{-1}. \quad (9-18)$$

We require the transformed data $\mathbf{X}_* = \mathbf{P}\mathbf{X}$, not the original data $\mathbf{X}$, to be well behaved.⁷ Under the assumption in (9-1), the following hold:

The GLS estimator is asymptotically normally distributed, with mean $\boldsymbol{\beta}$ and sampling variance

$$\text{Var}[\hat{\boldsymbol{\beta}} | \mathbf{X}_*] = \sigma^2(\mathbf{X}_*'\mathbf{X}_*)^{-1} = \sigma^2(\mathbf{X}'\boldsymbol{\Omega}^{-1}\mathbf{X})^{-1}. \quad (9-19)$$

The GLS estimator $\hat{\boldsymbol{\beta}}$ is the minimum variance linear unbiased estimator in the generalized regression model. This statement follows by applying the Gauss–Markov theorem to the model in (9-17). The result in Theorem 9.5 is **Aitken's theorem** (1935), and $\hat{\boldsymbol{\beta}}$ is sometimes called the Aitken estimator. This broad result includes the Gauss–Markov theorem as a special case when $\boldsymbol{\Omega} = \mathbf{I}$.

Hypothesis Testing

To test hypotheses, we can apply the full set of results we obtained before to this transformed model.

Important to notice, however, that there is no precise counterpart to the R^2 in the generalized regression model.

The GLS estimator minimizes the generalized sum of squares, ε_* . There is no assurance that dropping/adding variables would result in changes in the R^2 based on $\hat{\beta}$.

The transformed regression is a computational device, not the model of interest.

Feasible Generalized Least Squares

To use the results established above, $\mathbf{\Omega}$ must be known.

If $\mathbf{\Omega}$ contains unknown parameters that need to be estimated, the GLS is not **feasible**.

But with unrestricted $\mathbf{\Omega}$, there are far too many parameters to estimate with n observations.

The typical problem involves a small set of parameters $\boldsymbol{\alpha}$ such that $\mathbf{\Omega} = \mathbf{\Omega}(\boldsymbol{\alpha})$

Feasible Generalized Least Squares

A simple heteroskedasticity model would be

$$\sigma_i^2 = \sigma^2 z_i^\theta$$

for some exogenous variable z .

Suppose that $\hat{\alpha}$ is a consistent estimator of α . To make GLS feasible, we shall use $\hat{\mathbf{\Omega}} = \mathbf{\Omega}(\hat{\alpha})$

The FGLS estimator is

$$\hat{\hat{\beta}} = (\mathbf{X}'\hat{\mathbf{\Omega}}^{-1}\mathbf{X})^{-1}\mathbf{X}'\hat{\mathbf{\Omega}}^{-1}\mathbf{y}$$

Under some conditions, $\hat{\hat{\beta}}$ is asymptotically equivalent to $\hat{\beta}$ and the FGLS estimator based on $\hat{\alpha}$ has the same **asymptotic** properties as the GLS estimator.

Heteroskedasticity and the Weighted Least Squares

The GLS estimator is

$$\hat{\beta} = (\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{\Omega}^{-1}\mathbf{y}$$

Under Heteroskedasticity, $\mathbf{\Omega}^{-1}$ is a diagonal matrix whose i th diagonal element is $1/\omega_i$.

The GLS estimator is obtained by regression

$$\mathbf{Py} = \begin{bmatrix} y_1/\sqrt{\omega_1} \\ y_2/\sqrt{\omega_2} \\ \vdots \\ y_n/\sqrt{\omega_n} \end{bmatrix} \text{ on } \mathbf{PX} = \begin{bmatrix} \mathbf{x}'_1/\sqrt{\omega_1} \\ \mathbf{x}'_2/\sqrt{\omega_2} \\ \vdots \\ \mathbf{x}'_n/\sqrt{\omega_n} \end{bmatrix}$$

Heteroskedasticity and the Weighted Least Squares

Applying OLS to the transformed model, we obtain the **weighted least squares** estimator (WLS)

$$\hat{\beta} = \left[\sum_{i=1}^n (1/\omega_i) \mathbf{x}_i \mathbf{x}_i' \right]^{-1} \left[\sum_{i=1}^n (1/\omega_i) \mathbf{x}_i \mathbf{y}_i \right]$$

The logic of the computation is that observations with smaller variances receive a larger weight in the computations of the sums and therefore have greater influence in the estimates obtained.

Weighted Least Squares with *known* Ω

A common specification is that the variance is proportional to the one of the regressors or its square.

- For example, in a regression of family expenditures, the relevant variable is usually income.

If we assume $\sigma_i^2 = \sigma^2 x_{ik}^2$, the transformed regression model for GLS is

$$\frac{\mathbf{y}}{\mathbf{x}_k} = \beta_k + \beta_1 \frac{\mathbf{x}_1}{\mathbf{x}_k} + \beta_2 \frac{\mathbf{x}_2}{\mathbf{x}_k} + \dots + \beta_1 \frac{\varepsilon}{\mathbf{x}_k}$$

Weighted Least Squares with *known* Ω

If we know the structure of the disturbances variance, we can recover efficiency by adjusting the data to incorporate this information.

However, it is rarely possible to be certain about the nature of the heteroskedasticity in a regression model.

Using the wrong set of weights has two important consequences:

1. WLS becomes inefficient
2. the standard errors will be incorrect

The standard approach in the literature is to use OLS with the White estimator or some variant for the asymptotic covariance matrix.

Testing for Heteroskedasticity

Tests for heteroskedasticity will, in general, be applied to the OLS residuals (why?)

The LS estimator of β , in the presence of heteroskedasticity, although inefficient, is consistent.

Thus, statistics computed using the LS residuals $e_i = (y_i - \mathbf{x}_i' \mathbf{b})$ will have the same asymptotic properties as those computed using the true disturbances $\varepsilon = (y_i - \mathbf{x}_i' \beta)$.

White's General Test

White (1960) proposes a general hypothesis of the form

$$H_0 : \sigma_i^2 = E[\varepsilon_i^2 | \mathbf{X}] = \sigma^2 \text{ for all } i$$

$$H_1 : \text{Not } H_0$$

An approach is to use an F test to test the hypothesis that $\gamma_1 = 0$ and $\gamma_2 = 0$ in the regression

$$e_i^2 = \gamma_0 + \mathbf{X}_i' \gamma_1 + (\mathbf{x}_i \times \mathbf{x}_i') \gamma_2 + \nu_i$$

This test is extremely general. To carry it out, we need not make any specific assumptions about the nature of the heteroskedasticity.

The Breusch-Pagan Test

The Breusch-Pagan Test considers $E[\varepsilon^2|\mathbf{X}]$ to be a linear function of the explanatory variables:

$$E[\varepsilon^2|\mathbf{X}] = \delta_0 + \delta_1 x_1 + \dots + \delta_k x_k + \nu$$

Therefore, the null hypothesis of homoskedasticity is

$$H_0 : \delta_1 = \delta_2 = \dots = \delta_k = 0$$

$$H_1 : \text{At least of } \delta_j \text{ is not equal to } 0$$

Assuming the classical assumptions, H_0 can be tested using the usual F test for the joint significance of all explanatory variables.

Notice that we are relying on large sample properties since $\varepsilon^2 \geq 0$ cannot be normally distributed.

The Breusch-Pagan Test

The equation we estimate is

$$e_i^2 = \delta_0 + \delta_1 x_{1i} + \dots + \delta_k x_{ki} + \varepsilon'_i$$

The steps in performing the test are the following:

1. Estimate $\mathbf{y} = \mathbf{x}\beta + \varepsilon$ and find e
2. Estimate $\mathbf{e}'\mathbf{e} = \mathbf{x}\delta + \varepsilon'$
3. Perform the F test for $\delta = 0$

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